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$$\lim_{x\to\infty}(\sqrt{x^2-1}-x)$$
$$= \lim_{x\to+\infty}(\sqrt{x^2-1}-x)\cdot\frac{\sqrt{x^2-1}+x}{\sqrt{x^2-1}+x}$$
$$= \lim_{x\to+\infty}\frac{(x^2-1)-x^2}{\sqrt{x^2-1}+x}$$
$$= \lim_{x\to+\infty}\frac{-1}{\sqrt{x^2-1}+x}$$
$$= \frac{-1}{+\infty+\infty} = 0$$
$$\lim_{x\to-\infty}(\sqrt{x^2-1}-x) = +\infty+\infty = +\infty$$

References